

Examining the Effectiveness of Obfuscatory Planning Strategies through Human Observation



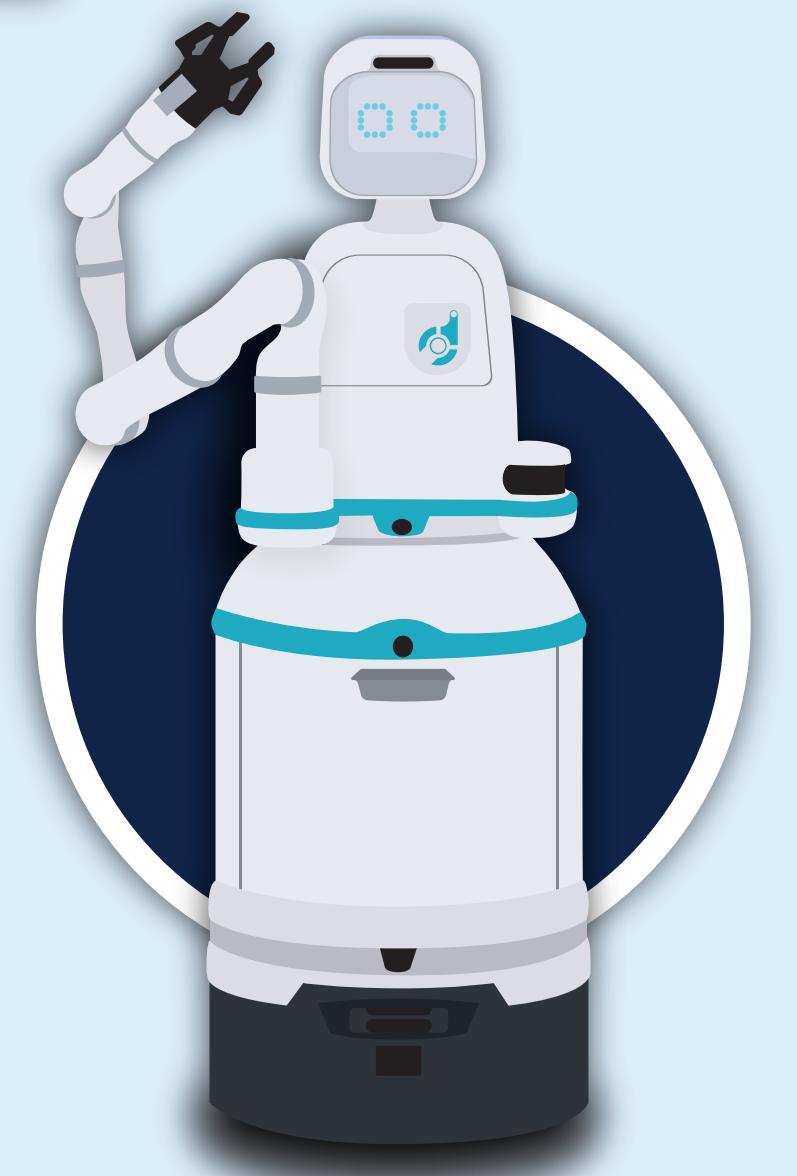
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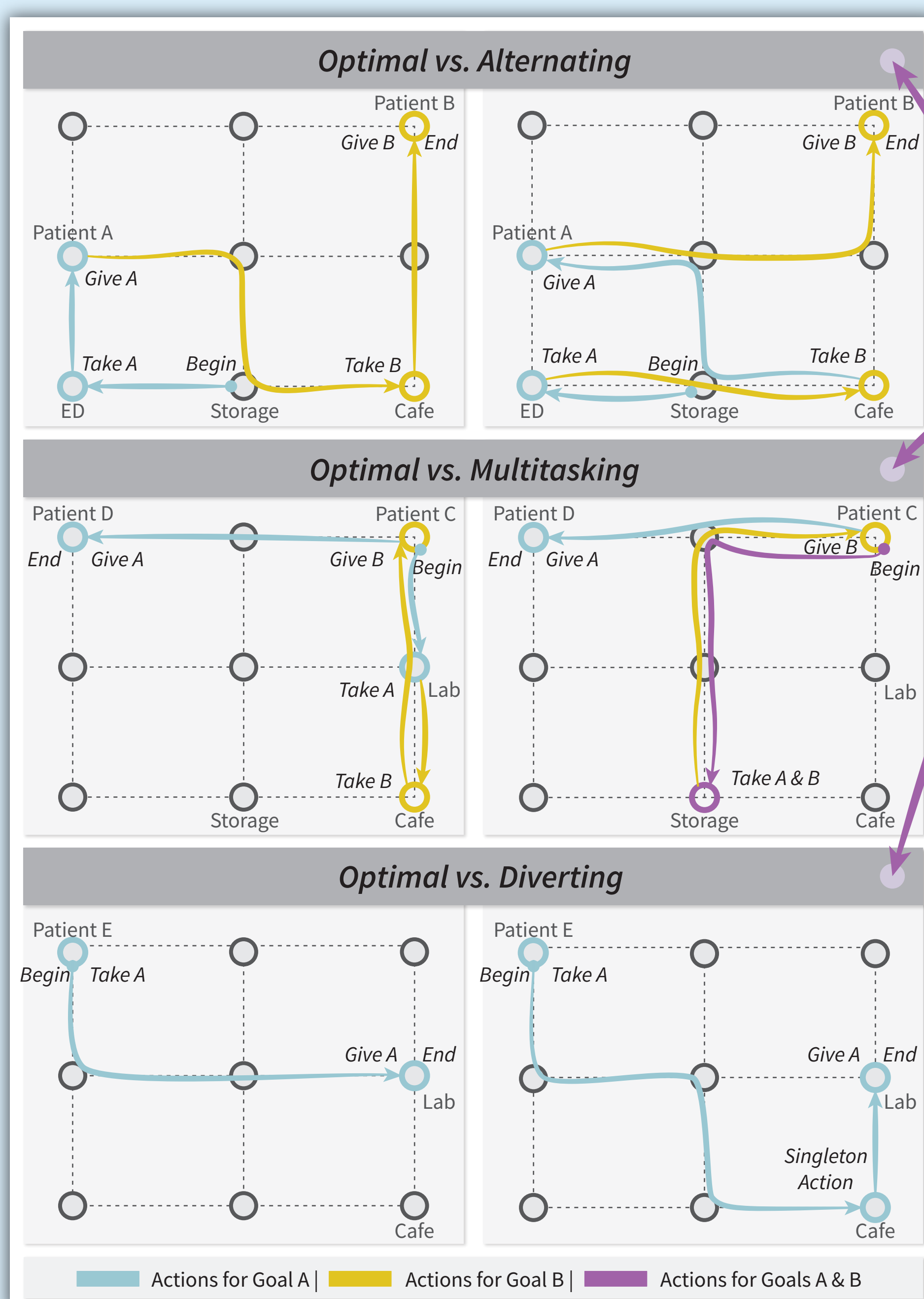
Obfuscatory Planning for Improved Privacy

Overview: As robots increasingly enter into public and private environments, we gain a better understanding of the risks associated with their day-to-day use. One such risk is **inference-based privacy leakage**, where observers infer private information from public signals. Here we utilize obfuscatory planning to conceal the goals of a simulated robot as it completes hospital deliveries.



Approach

Obfuscatory Planning & Human Observation



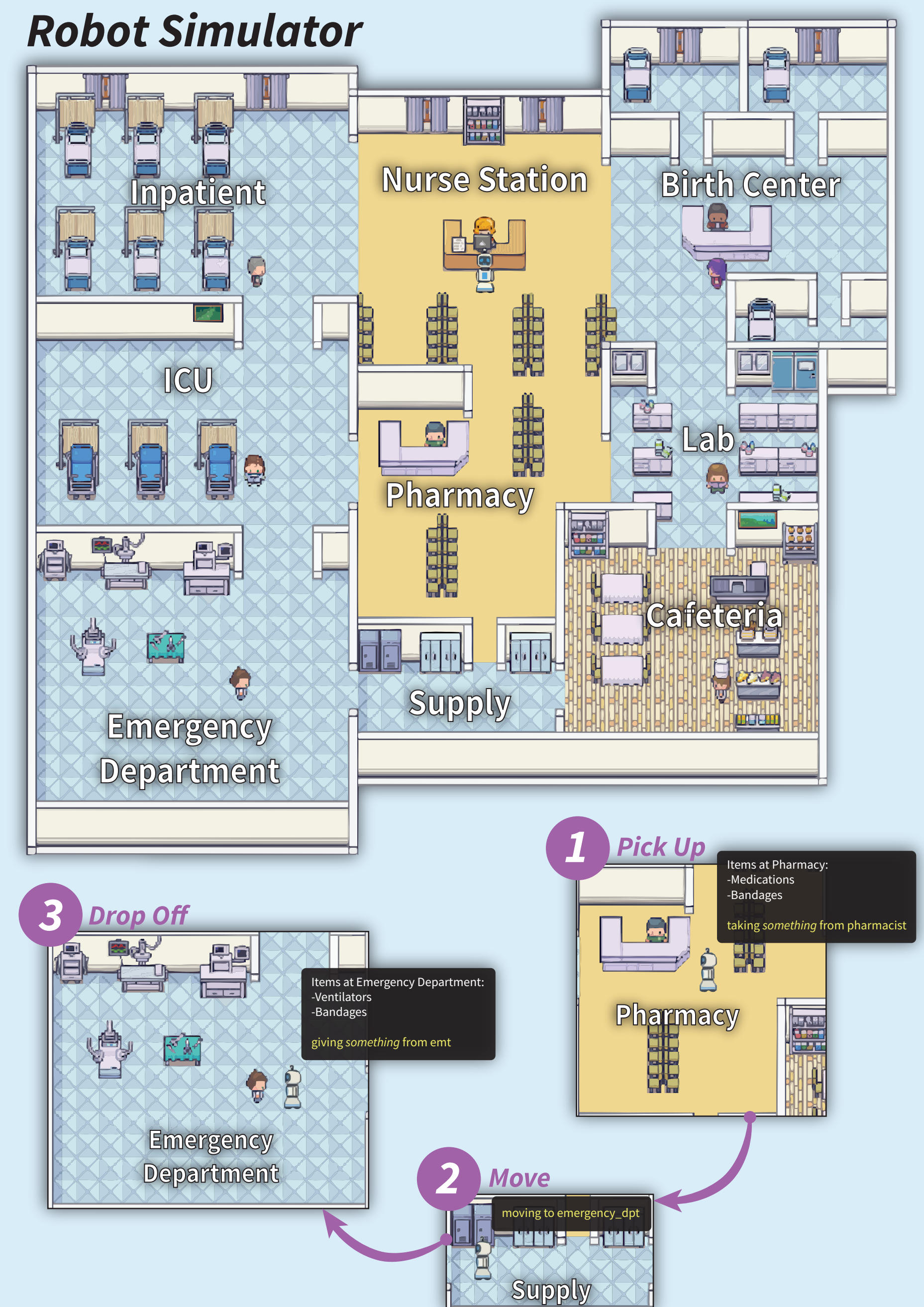
We developed three obfuscatory planning strategies to conceal a robot's goals. These strategies include:

- The **alternating** strategy interleaves the actions required to satisfy multiple goals.
- The **multitasking** strategy generates plans that encourage actions that progress multiple goals.
- The **diverting** strategy adds extraneous actions to the plan, increasing the set of goals the robot appears to be achieving.

We evaluated our developed strategies through a **large online study (N = 160)**. This study utilized a **simulated hospital environment** where a robot moves between rooms to complete delivery tasks.

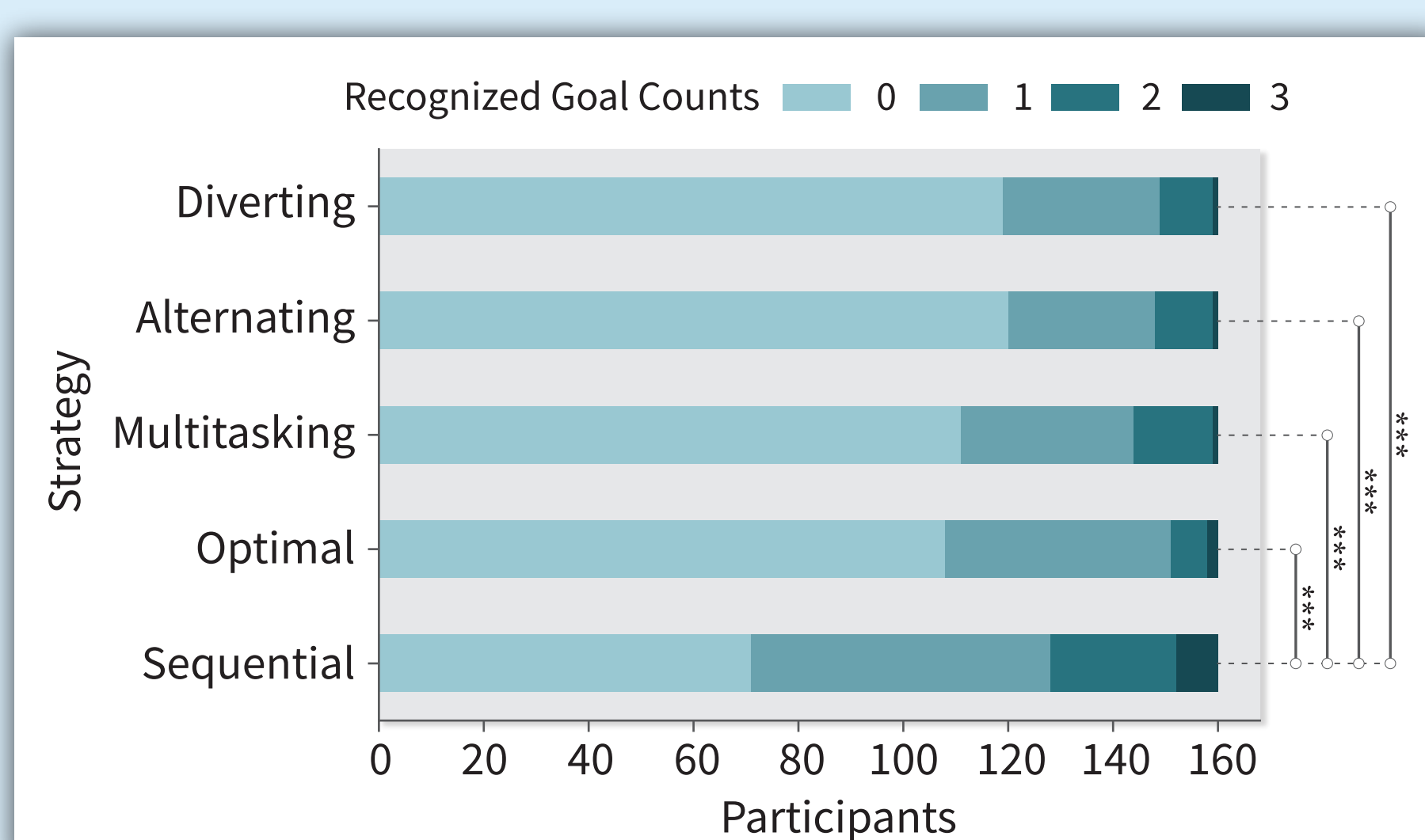
Participants were asked to identify the robot's goals under each obfuscatory strategy and two baseline strategies (i.e., optimal and sequential). A simple example delivery is provided (right).

Robot Simulator



Findings

Obfuscation through Optimal Planning



Participants generally **struggled to identify** the robot's goals (left) while simultaneously reporting **high confidence in their ability** to do so (right), regardless of the planning strategy.

These results suggest that **optimal planning may be sufficient** to obfuscate a robot's goals. This work illustrates the importance of human evaluation in obfuscatory planning to fully understand strategy performance.

